

Joan Pascual Ribes

Final-year student, BSc Mathematics + BSc Engineering Physics (CFIS, UPC).
Barcelona, Spain | jpascualribes@gmail.com
github.com/JoanPascualRibes | linkedin.com/in/joan-pascual-ribes-7a974b284 | joanpascualribes.github.io

Research experience in applied algebraic geometry and algebraic statistics (two Max Planck internships); currently training in the mathematical foundations of AI alignment.

EDUCATION

2023 – 2028	BSc Mathematics + BSc Engineering Physics Universitat Politècnica de Catalunya (UPC), Barcelona Double degree within CFIS, UPC's selective programme for high-performing students. Expected graduation: January 2028. · GPA 9.25/10; honours (<i>Matrícula d'Honor</i>) in 14 courses across both degrees. University entrance exam (PAU): 13.832/14. · Full tuition and residence scholarship for the whole double degree, granted by CFIS.
Sep – Oct 2026	Iliad Intensive — Foundational AI Alignment Iliad, London, UK Four-week, full-time course on mathematical approaches to AI safety: singular learning theory, deep learning theory, agent foundations, interpretability. Selective admission on mathematical strength.

RESEARCH EXPERIENCE

Aug – Sep 2026	Research Intern Max Planck Institute for Mathematics in the Sciences, Dresden Algebraic statistics applied to game theory, supervised by Dr. Irem Portakal.
Jun – Aug 2025	Research Intern Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG), Dresden Applied algebraic geometry in Dr. Aida Maraj's group: Lie-theoretic methods to characterise properties of ideals and varieties. Resulting paper in preparation.
Jul 2024	Barcelona Introduction to Mathematical Research (BIMR) Centre de Recerca Matemàtica (CRM) One-month research programme (UAB/UB/UPC) in algebra: representation of polynomials as sums of squares.

TEACHING AND PROFESSIONAL EXPERIENCE

Sep 2025 – Jan 2026	Teaching Assistant, Computer Science (Informàtica) BSc Mathematics, UPC Supported lectures and lab sessions and assisted students throughout the course (C++ and Python).
Mar – May 2025 Sep – Oct 2025	Prompt Engineer and AI Tutor (freelance) Outlier AI Evaluated and refined language-model outputs on advanced mathematical tasks; designed prompts and graded reasoning used for model training.

PROJECTS

Computational physics	Simulation of superfluid helium-II Team project. Spectral-method simulations reproducing the behaviour of the He-II superfluid. Repository .
Quantitative finance	Paper trader for Polymarket Team project. Python paper-trading engine for backtesting quantitative strategies on prediction markets.

OLYMPIADS AND DISTINCTIONS

Olympiads	Physics: 1st place (team), national phase of the <i>PlanckS</i> olympiad (Spain); selected to represent Spain at the international <i>PlanckS</i> competition (2026), placing 14th of ~60 teams. Mathematics: winner, Cangur Olympiad (Valencia), 2023, final-year high-school category.
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SKILLS

Programming	Python (NumPy, SciPy: numerical simulation, data analysis, backtesting), C++, SageMath, Macaulay2, $\LaTeX$, Git, Linux, Claude Code.
Languages	Spanish (native) · Catalan/Valencian (native, C2) · English (C1, Trinity College London) · German (B1, in progress) · French (A2).

INTERESTS AND VOLUNTEERING

Interests	Applied mathematics (optimisation, differential equations, numerical analysis and their applications in physics, engineering and economics); trading and markets, through their connection to probability, statistics and game theory; data science, AI and machine learning in their relation to mathematics.
Other	Mathematics teacher at Càritas (academic support for under-resourced students); saxophone and piano at conservatory level; poetry and essays.